

Karen Xu

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EDUCATION

University of Victoria

September 2021 – May 2026

Bachelor of Science in Computer Science

Victoria, BC

- Graduated with Distinction, Dean's List (4.0 GPA, Top 10% in the Faculty of Engineering)
- Relevant Courses: Data Mining, Systems for Massive Datasets, Data Structures & Algorithms

EXPERIENCE

Research Assistant

January 2026 – September 2026

University of Victoria

Victoria, BC

- Awarded NSERC research funding to investigate graph-based retrieval-augmented generation (RAG) and evaluation methods in the biomedical domain
- Developed a full-stack web application that converts hundreds of biomedical papers into a queryable knowledge graph, generating answers and relevant sources given user queries
- Designed and ran evaluations to compare our novel retrieval method against standard baselines, showing 28% higher retrieval accuracy while using roughly 65x less memory
- Created a Python library that transforms research papers from multiple formats (XML, Markdown, PDF) into a standardized JSON schema, expanding our available dataset without losing consistency
- Reduced document processing time by 12x by partitioning papers into small batches and submitting them as independent jobs to a Linux HPC cluster

Automation Analyst Co-op

January 2025 – August 2025

Government of BC, Network Services

Victoria, BC

- Developed a data pipeline that produces office occupancy estimates from network activity logs, informing property management decisions contributing to over \$4M in savings
- Led regular consultations with stakeholders to gather requirements, delivering 5 projects over the 8-month co-op with a 100% adoption rate
- Created a network topology mapping system using provider APIs to automatically identify connected devices, replacing the need for manual mapping
- Demonstrated a viable network security alerting approach by training a neural network classifier to detect irregular traffic patterns, achieving 0.977 ROC-AUC after using SMOTE to account for class imbalance
- Extended a data pipeline that synthesized network device data from multiple service providers by implementing a conflict resolution mechanism, ensuring a single source of truth for each device

PROJECTS

Chef's Choice | *Python, FastAPI, PostgreSQL, React, SQLAlchemy, Alembic*

August 2026 – Present

- Created a data pipeline to normalize and deduplicate restaurant data from multiple sources before storing in a PostgreSQL database
- Created a RESTful API for efficiently querying, filtering, and sorting restaurant data
- Designed a user-friendly React dashboard that helps users find the best value restaurants in the area

Exoplanet AI, NASA SpaceApps Hackathon 2025 | *Python, PyTorch*

October 2025

- Trained a convolutional neural network to detect exoplanets using stellar light curve data from NASA
- Engineered predictive features by transforming and fitting noisy time-series data to splines
- Identified and addressed data quality issues including missing values, outliers, and noise to improve model performance

SKILLS

Programming Languages: Python, SQL, JavaScript, HTML, CSS

Frameworks & Libraries: React, PyTorch, FastAPI, SQLAlchemy, Alembic

Databases: PostgreSQL, Milvus (Vector), Memgraph (Graph)

Spoken Languages: English, French, Mandarin